NIST seed skip filter: new deterministic features

2026-10-17 · research/nist_seed_skip_filter_20261017 · preregistration commit 3586810ea

Status. Measured negative result. None of the 21 new preregistered binary features (H7–H11) explains why 11 of the 21 valid NIST/X9.62 seed candidates were not published. This concerns seed provenance, not ECDLP hardness.

Population. The 21 candidates accepted by every known criterion in the 2026-10-16 census (`../nist_seed_provenance_20261016`): 10 published, 11 skipped. Features are functions of (p, n), where $t = p + 1 - n$, $D = t^2 - 4p$ and $n' = 2p + 2 - n$.

Hypotheses. H7: $Z[\pi]$ is maximal at every odd $l < 10^6$. H8: the largest prime factor of the twist order (thresholds 160 bits and $\text{bitlen}(p) - 32$; cofactor of at most 32 bits). H9: the same thresholds for $n \pm 1$. H10: a shortcut MOV test, $\left(\frac{p}{n}\right)$, $\left(\frac{-p}{n}\right)$, whether p is a primitive root mod n, and whether $\frac{n-1}{\text{ord}_{n(p)}} \leq 2$. H11: $\left(\frac{x}{n}\right)$ for $x \in \{-1, 2, 3, t, n'\}$ and $\left(\frac{y}{p}\right)$ for $y \in \{n, t, D, n'\}$.

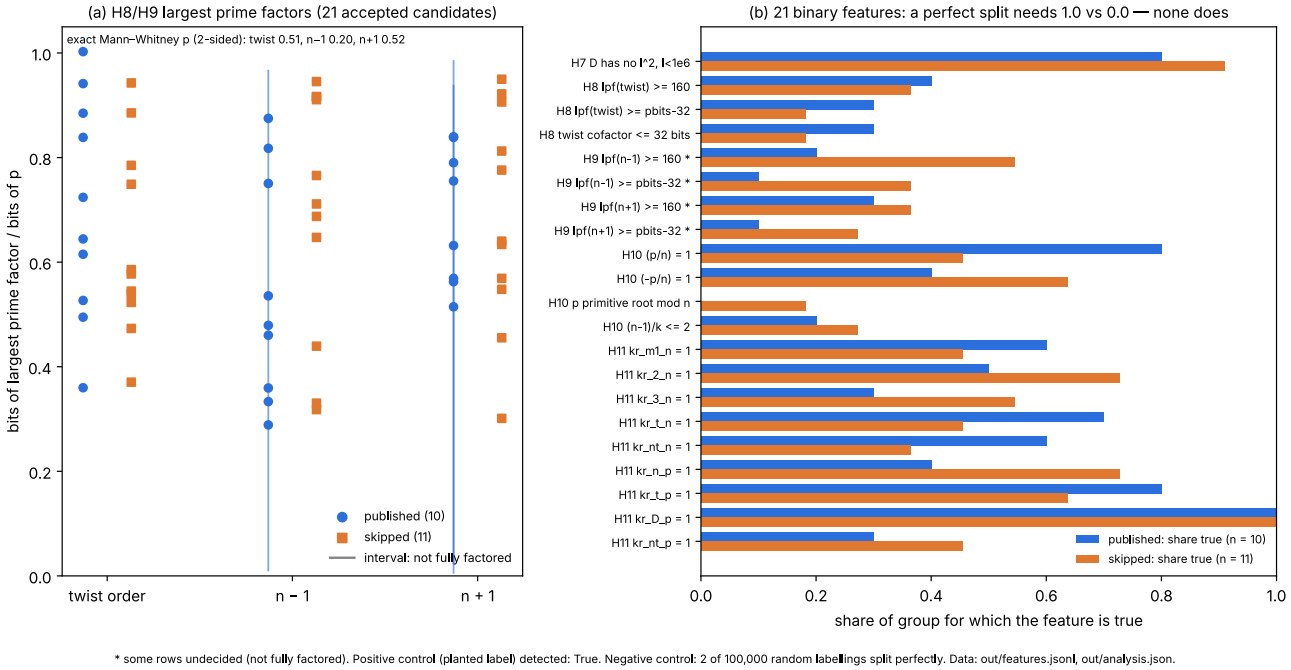


Figure 1: Recorded values (`out/features.jsonl`, `out/analysis.json`). (a) Largest prime factor of the twist order and of $n \pm 1$, in bits relative to p; the lines are intervals where factoring did not finish in 600 s. (b) For each binary feature, the share of each group for which it is true. A perfect split would be 1.0 against 0.0.

Results (measured). No binary feature splits the groups perfectly. Strongest leans (partial, not claimed): $\left(\frac{p}{n}\right) = 1$ in 8/10 published and 5/11 skipped; $\text{lpf}(n-1) \geq 160$ in 2/10 and 6/11. Exact two-sided Mann–Whitney tests over all $\binom{21}{10}$ labellings: twist $p = 0.51$, $n - 1$ $p = 0.20$, $n + 1$ $p = 0.52$, count of square factors of D $p = 0.59$. Positive control (planted label) detected. Negative control: 2 of 100,000 random labellings split perfectly on some feature (family-wise rate $2 \cdot 10^{-5}$). Unfactored after 600 s: P-521 $n + 1$, P-384 $n \pm 1$; all three are on published curves, which can only block a split. Cross-checks: the largest prime factor of the twist order is 241 bits for P-256 and 118 bits for P-224, both matching known values.

Caveats. $\left(\frac{D}{p}\right) = 1$ holds by construction ($D \equiv t^2$) and was included by mistake. Two H10 conditions are the same condition and are reported once. A bug that inverted the "fully factored" flag was fixed, and the run repeated, before any analysis.

What remains. About 80 deterministic features have now been ruled out across both studies. The data fit two hypotheses that exact computation cannot test. One is a randomised check drawn from a generator outside the seed stream (for example, a random-point test $[n]P = O$ without a square-root check, which half of all curves fail). The other is a manual choice. The B-163 seed and four of the base points are still unexplained.